

Spanning Question Test Paper

1. What is the acceleration due to gravity on Earth?

- (A) 9.8 m/s^2
- (B) 8.9 m/s^2
- (C) 10.8 m/s^2
- (D) 7.8 m/s^2

2. A block of mass 10 kg slides down a frictionless inclined plane of angle 30 degrees.

If the plane has a total length of 20 meters, determine the time taken by the block to reach the bottom starting from rest, and choose the correct answer below:

- (A) 2.02 seconds
- (B) 2.86 seconds

- (C) 3.50 seconds
- (D) 4.10 seconds

3. What is the unit of magnetic flux density?

- (A) Tesla
- (B) Weber
- (C) Henry
- (D) Gauss

4. Consider a thermodynamic heat engine operating between two thermal reservoirs at temperatures $T_{\text{hot}} = 600 \text{ K}$ and $T_{\text{cold}} = 300 \text{ K}$. The working substance undergoes a Carnot cycle consisting of two reversible isothermal processes and two reversible adiabatic processes.

During the isothermal expansion at 600 K, the engine absorbs 1200 Joules of heat from the hot reservoir.
Calculate the theoretical maximum efficiency and the net mechanical work output performed per cycle:

(A) Efficiency = 50%, Work = 600 J

(B) Efficiency = 40%, Work = 480 J

(C) Efficiency = 60%, Work = 720 J

(D) Efficiency = 75%, Work = 900 J

5. Which law of thermodynamics establishes the concept of entropy?

(A) Zeroth Law

(B) First Law

(C) Second Law

(D) Third Law